

THE UNIVERSITY OF AZAD JAMMU & KASHMIR
MUZAFFARABAD

DATABASE MANAGEMENT SYSTEM

Lab 3 — Keys and Queries

University Database System

Name	Tayyab Hanif Awan
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Submitted To	Engr. Awais Rathore
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1. Objective

- Identify and implement Primary, Foreign, Composite, Candidate, Alternate, Super, Natural, and Surrogate keys.
- Create a multi-table university schema with correct relationships and constraints.
- Retrieve data using SELECT and JOIN queries.
- Modify an existing schema using ALTER TABLE (add, modify, rename, drop columns and constraints).
- Perform UPDATE, DELETE, TRUNCATE, and DROP operations while respecting referential integrity.

2. Database Structure

The database (keys_and_queries_db) models a small university system with 5 tables.

2.1 Tables

Table	Purpose
departments	Surrogate PK (dept_id); dept_name is a Natural/Alternate Key
students	Surrogate PK (student_id); national_id & email are Candidate/Alternate Keys
courses	Surrogate PK (course_id); course_code is a Natural/Alternate Key
instructors	Surrogate PK (instructor_id); email is an Alternate Key
enrollments	Composite Primary Key (student_id + course_id); junction table

3. Key Tasks / Concepts Implemented

- Explaining and demonstrating each key type using the schema above.
- Inserting sample data into all 5 tables.
- Writing SELECT and JOIN queries (e.g. student-department mapping, enrollment reports).
- ALTER TABLE: adding, modifying, renaming, and dropping columns and constraints.
- UPDATE and DELETE operations, respecting foreign key dependencies (child rows before parent rows).
- TRUNCATE TABLE and DROP TABLE / DROP DATABASE for full cleanup.

4. Implementation (phpMyAdmin)

All key types and CRUD/DDL operations were executed and verified using phpMyAdmin under XAMPP.

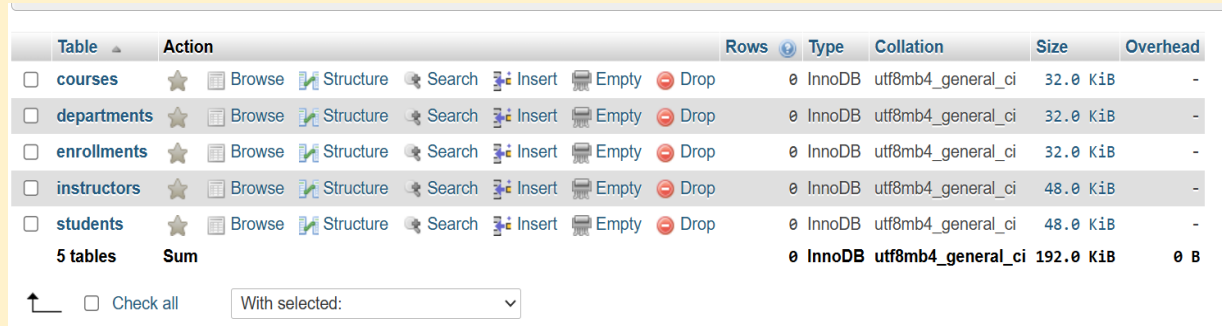
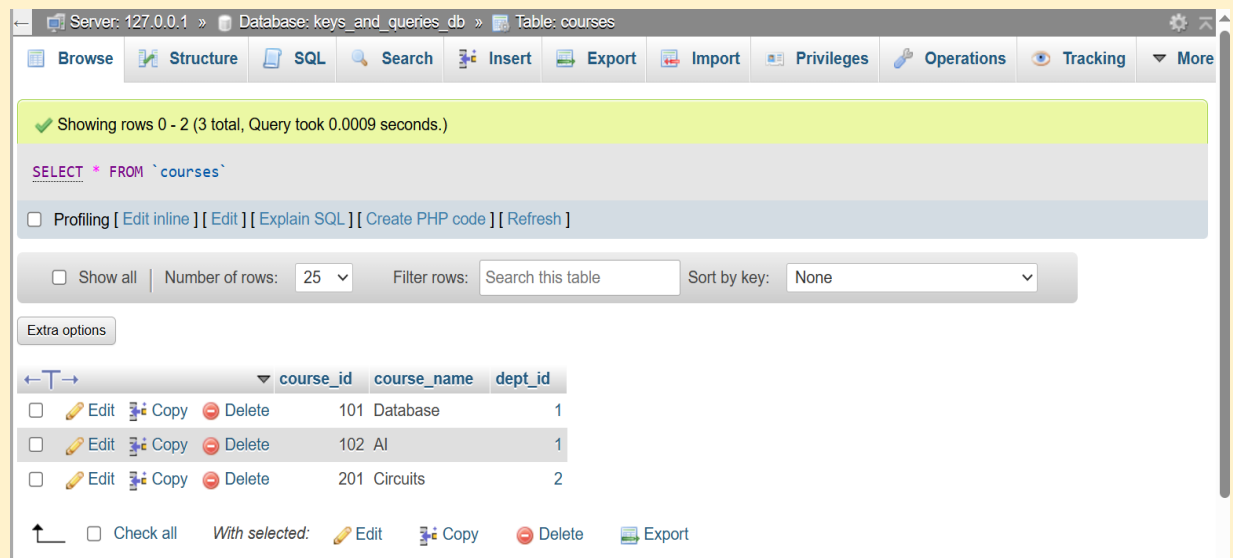


Table	Action	Rows	Type	Collation	Size	Overhead
☐ courses	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ departments	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ enrollments	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ instructors	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	48.0 KiB	-
☐ students	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	48.0 KiB	-
5 tables	Sum	0	InnoDB	utf8mb4_general_ci	192.0 KiB	0 B

☐ Check all With selected: ▾

Structure tab showing 5 tables



Server: 127.0.0.1 » Database: keys_and_queries_db » Table: courses

Browse Structure SQL Search Insert Export Import Privileges Operations Tracking More

✓ Showing rows 0 - 2 (3 total, Query took 0.0009 seconds.)

`SELECT * FROM `courses``

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	course_id	course_name	dept_id
☐ Edit Copy Delete	101	Database	1
☐ Edit Copy Delete	102	AI	1
☐ Edit Copy Delete	201	Circuits	2

☐ Check all With selected: Edit Copy Delete Export

Browse tab (showing sample data for courses table)

✓ Showing rows 0 - 1 (2 total, Query took 0.0009 seconds.) [student_id: 1... - 2...]

```
-- 8.2 Students with their departments SELECT s.student_id, s.name AS student_name, d.dept_id, d.dept_name FROM students AS s INNER JOIN departments AS d ON s.dept_id = d.dept_id ORDER BY s.student_id;
```

[Edit inline] [Edit] [Create PHP code]

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

student_id	student_name	dept_id	dept_name
1	Ali Khan	1	CS
2	Sara	1	CS

☐ Show all | Number of rows: 25 | Filter rows: Search this table

JOIN (students with department name)

Extra options

	student_id	name	email	age	dept_id	phone
☐ Edit Copy Delete	1	Ali Khan	ali@gmail.com	20	1	NULL
☐ Edit Copy Delete	2	Sara	sara@gmail.com	21	1	NULL

☐ Check all
 With selected:
 Edit Copy Delete Export

Phone is added in student table after ALTER query

5. Conclusion

This lab reinforced the theory behind every major type of relational key by implementing each one in a real schema, alongside practical schema evolution and safe data mutation.